

GNSS-BASED CRUSTAL DEFORMATION AND STRAIN ACCUMULATION IN THE NEPAL HIMALAYA WITH INTEGRATED WEB-BASED VISUALIZATION

¹Dipendra Kumar Thakur, ²Bhuwan Singh Bishta

¹Department of Geomatics Engineering, Western Regional Campus, Tribhuvan University, Nepal.

²Land Management Training Center, Dhulikhel, Nepal.

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ABSTRACT

Plate motion in the Nepal Himalaya is primarily driven by convergence between the Indian and Eurasian plates along a series of large-scale thrust faults with clear surface expressions. Understanding present-day crustal deformation patterns is essential for assessing seismic hazard in this tectonically active region. This study investigates the rates and spatial characteristics of crustal deformation across the Nepal Himalaya using GNSS and geodetic survey data integrated within a web-based analytical framework. Continuous and campaign GNSS observations from regional and international geodetic networks were analyzed to estimate horizontal and vertical crustal velocities. Station velocity fields were derived from long-term position time series, and crustal strain rates were computed using spatial velocity gradient techniques.

The results reveal pronounced northward and eastward crustal motion across Nepal, with northward velocities of approximately 15–20 mm/yr in central Nepal, consistent with interseismic locking along the Main Himalayan Thrust. Elevated compressional strain rates are observed in the Himalayan frontal region, indicating significant accumulation of elastic strain. Limited relative motion across the frontal thrust system suggests partial fault locking and sustained tectonic loading. In addition, this study develops a web-based GNSS data analysis and visualization platform that enables interactive exploration of velocity vectors, displacement time series, strain-rate distributions, and regional seismicity through a browser-based interface. By combining established geodetic methodologies with modern web technologies, this approach enhances accessibility, transparency, and reproducibility of crustal deformation analysis in the Nepal Himalaya.

1. INTRODUCTION

Surveying using the Global Navigation Satellite System (GNSS) has proven to be one of the most reliable geodetic techniques for investigating geophysical processes associated with earthquakes and crustal deformation (Blewitt & Lavallée, 2002; Segall, 2010). The development of the Global Positioning System (GPS), with millimeter-level positional accuracy, has significantly enhanced the capability to monitor slow and continuous movements of the Earth's surface. Such precise measurements have provided new insights into the processes through which stress and strain accumulate within the Earth's crust prior to earthquake occurrence.

Crustal deformation is a gradual process that oper-

ates over both short-term and long-term timescales. Continuous GNSS monitoring is particularly effective for observing the progressive accumulation of tectonic strain in near real time. GNSS data enable quantification of long-term deformation rates, thereby improving understanding of stress build-up and release mechanisms within active tectonic regions. When combined with regional seismicity data, geodetic observations contribute to more accurate identification of earthquake-prone zones and improved seismic hazard assessment.

Numerous studies have characterized deformation of the Indian plate using GNSS observations (Banerjee et al., 2008; Ader et al., 2012). These investigations consistently indicate that the Indian plate is moving northeastward at a rate of approximate-

ly 50–55 mm/yr (Paul et al., 2001; Banerjee et al., 2008; Ader et al., 2012). Nearly half of this convergence is accommodated within the Himalayan thrust and fold belt, while the remaining motion is transferred northward into Tibet and Central Asia. This ongoing convergence plays a fundamental role in generating large and destructive earthquakes in the Himalayan region.

Previous studies of major Himalayan fault systems have reported variations in shortening and slip rates along structures such as the Main Boundary Thrust, Main Frontal Thrust, and Karakoram Fault (Bettinelli et al., 2006; Stevens & Avouac, 2015). Investigations across the western, central, and eastern Himalaya suggest that several fault segments remain partially or fully locked, leading to significant interseismic strain accumulation. In particular, geodetic evidence indicates that segments of the Himalayan Frontal Thrust in Nepal and the western Himalaya are largely locked, increasing the potential for future large-magnitude earthquakes.

The occurrence of recurrent destructive earthquakes in the Himalayan region underscores the importance of detailed crustal deformation studies. Despite valuable insights provided by earlier research (Bettinelli et al., 2006; Banerjee et al., 2008; Ader et al., 2012; Stevens & Avouac, 2015), notable discrepancies remain in reported slip rates, contraction estimates, and strain distribution patterns. These differences highlight the need to reassess crustal deformation using high-precision and updated GNSS observations. Accordingly, this study focuses on interpreting present-day crustal deformation in the Nepal Himalaya, with particular emphasis on the Himalayan Frontal Thrust, to improve understanding of strain accumulation and its implications for seismic hazard in the central and western Himalaya.

Although numerous GNSS-based studies have examined Himalayan tectonics, variations persist in estimates of convergence rates, strain accumulation, and fault coupling across Nepal. Furthermore, much of the existing work relies on static maps and offline processing environments, which limit data accessibility, reproducibility, and interactive analysis. This study aims to quantify present-day crustal deformation and strain accumulation in the Nepal Himalaya using continuous and campaign GNSS observations and to visualize these processes through an interactive web-based system. Specifically, this research estimates horizontal and

vertical GNSS velocity fields, analyzes long-term GNSS position time series to identify deformation trends, computes crustal strain rates to delineate zones of high compression, and integrates deformation patterns with regional seismicity. By combining geodetic analysis with an interactive visualization framework, this study enhances accessibility, transparency, and interpretation of GNSS-derived crustal deformation information relevant to seismic hazard assessment in the Nepal Himalaya.

2. MATERIALS AND METHODS

2.1 Study Area

The study region covers the central, western, and eastern Nepal Himalaya, extending approximately between 80°E–88°E longitude and 26°N–30°N latitude as shown in Figure 1. Nepal lies along the active convergent boundary between the northward-moving Indian Plate and the relatively stable Eurasian Plate and exhibits extreme topographic variation, ranging from lowland plains in the south to the world's highest mountain ranges in the north. The region spans the Himalayan foreland, Lesser Himalaya, and Higher Himalaya and is structurally controlled by major east–west–trending fault systems, including the Main Frontal Thrust, Main Boundary Thrust, and Main Himalayan Thrust, which govern regional deformation and mountain-building processes.

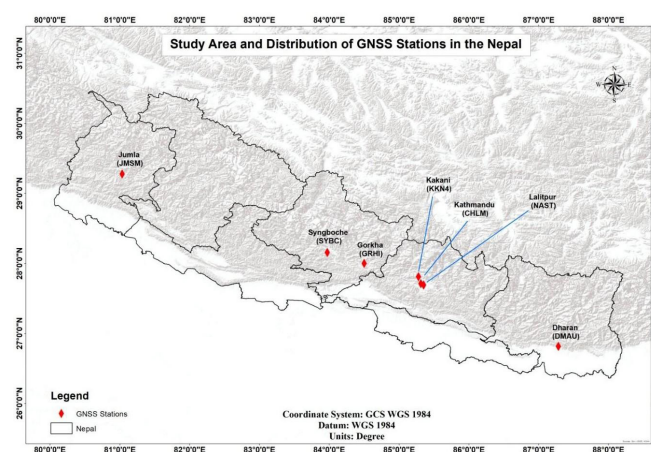


Figure 1:- Study area of the Nepal Himalaya showing topography, major tectonic structures, and GNSS station distribution.

2.2 Methodology

This study employs a combined geodetic and web-based analytical approach to estimate and map crustal deformation rates across the Nepal Himalaya using GNSS and geodetic survey data. The methodology follows established scientific procedures and emphasizes transparency, reproducibility, and accessibility through a browser-based platform. The overall methodological workflow is illustrated in Figure 2.

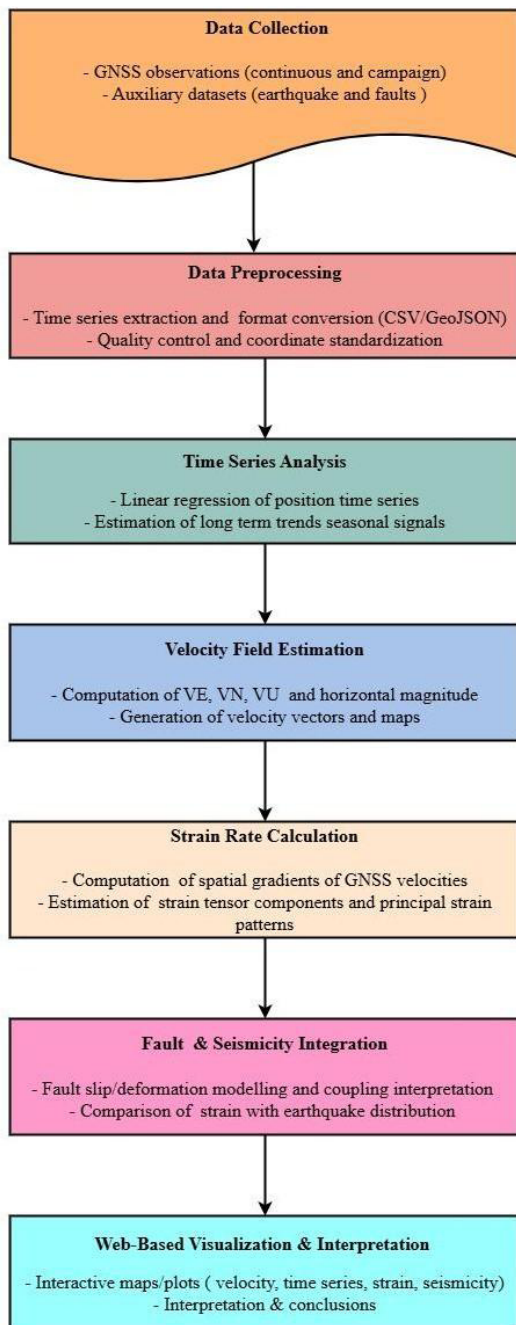


Figure 2:-Working Methodology.

2.3 Data Collection

This study is based on GPS and GNSS observations obtained from globally recognized repositories, primarily the Nevada Geodetic Laboratory and the UNAVCO data archives. These sources provide long-term GNSS position time series and pre-processed velocity data from stations distributed across Nepal and the surrounding Himalayan region. Table 1 shows selected stations CHLM, KKN4, JMSM, NAST, SYBC, DMAU, and GRHI ensure representative spatial coverage from the Himalayan foreland to the Higher Himalaya.

Continuous and campaign GNSS observations spanning approximately 2010 to 2023 were used, providing sufficient temporal coverage to estimate reliable long-term crustal deformation rates.

Table 1:- GNSS stations, Locations and observation periods.

Station Name	Location	Observation Period
CHLM	Kathmandu	2010-2023
KKN4	Kakani	2011-2023
JMSM	Jumla	2012-2023
NAST	Lalitpur	2010-2023
SYBC	Syangboche	2013-2023
DMAU	Dharan	2011-2023
GRHI	Gorkha	2012-2023

To support seismic interpretation, earthquake catalogs were obtained using the web-based application programming interface (API) of the United States Geological Survey placed in Table 2. The dataset includes earthquake location, magnitude, depth, and occurrence time for events with magnitudes ≥ 4.0 in the Nepal region. Fault line data for major tectonic structures, including the Main Himalayan Thrust, Main Boundary Thrust, and Main Frontal Thrust, were incorporated from global fault databases and national geological survey records to provide tectonic context.

Table 2:- Data Sources and Uses

SN	Item	Source	Uses
1	GNSS station time series	Nevada Geodetic Laboratory (NGL) GPS Time Series Data https://geodesy.unr.edu/gps_timeseries/	Velocity and time-series analysis
2	GNSS velocity fields	Nevada Geodetic Laboratory (NGL) GPS Velocity Fields https://geodesy.unr.edu/index.php (navigate to velocity products)	Regional deformation mapping
3	Earthquake catalog	U.S. Geological Survey (USGS) Earthquake Catalog https://earthquake.usgs.gov/earthquakes/search/	Seismicity correlation
4	Fault line data	GEM Global Active Faults Database (Global Earthquake Model) https://hub.arcgis.com/datasets/37a384d4c1ef-4f56a33a40f291a634e9_0	Tectonic interpretation

2.4 Preparation and Preprocessing of Data

Raw GNSS data were converted into user-friendly and web-compatible formats (CSV and GeoJSON). The GNSS velocity datasets include east, north, and vertical components along with their associated uncertainties. GNSS position time series were organized as daily displacements relative to a reference epoch. All datasets were examined for incompleteness, inconsistencies, and outliers, and appropriate quality control procedures were applied prior to integration into the web-based analysis system. Earthquake and fault datasets were spatially clipped to the Nepal Himalaya to improve analytical relevance and computational efficiency. All datasets were standardized to a common geographic coordinate reference system to ensure consistency in spatial analysis and visualization.

2.5 Time Series Analysis

Long-term station velocities were determined using GNSS position time series data. Annual velocity rates were estimated through linear regression applied to the displacement time series in the north, east, and vertical components. The displacement model is expressed as:

$$X(t) = vt + c + A_1 \sin(2\pi t) + B_1 \cos(2\pi t) + A_2 \sin(4\pi t) + B_2 \cos(4\pi t) + \varepsilon(t)$$

where v represents the station velocity, c is the constant offset corresponding to the station position at the reference epoch and is estimated using least-squares linear regression, A_1 and B_1 are the coefficients of the annual periodic signal, A_2 and B_2 are the coefficients of the semi-annual periodic signal, and ε denotes the residual noise (Blewitt & Lavallée, 2002). Outliers resulting from equipment changes, data gaps, or coseismic offsets were identified using statistical thresholds and removed from the time series. Seasonal signals associated with hydrological loading and monsoon effects were evaluated through visual inspection and trend analysis.

2.6 Velocity Field Estimation

Processed GNSS position time series were used to compute horizontal and vertical velocity vectors for each station. The horizontal velocity magnitude (V) was calculated as:

$$V = \sqrt{V_E^2 + V_N^2}$$

Where V_E and V_N represent the eastward and northward velocity components, respectively. The resulting velocity field represents present-day crustal motion and was analyzed to identify patterns of convergence, uplift, and lateral deformation across the study area. The estimated velocity field was compared with results from previously published geodetic studies to assess consistency and reliability.

2.7 Strain Rate Calculation

GNSS-derived velocity fields were used to calculate crustal strain rates based on spatial gradients of station velocities using established geodetic techniques. Velocity derivatives were computed in the east-west and north-south directions using groups of neighboring GNSS stations. From these derivatives, strain components representing extensional, compressional, and shear deformation were derived.

Crustal strain rates were calculated following standard geodetic formulations (Savage et al., 2001). The components of the two-dimensional strain tensor were computed as:

$$\varepsilon_{xx} = \frac{\partial v_x}{\partial x}, \quad \varepsilon_{yy} = \frac{\partial v_y}{\partial y}, \quad \varepsilon_{xy} = \frac{1}{2} \left(\frac{\partial v_x}{\partial y} + \frac{\partial v_y}{\partial x} \right)$$

Here ε_{xx} and ε_{yy} represent normal strain rates in east-west and north-south directions respectively and ε_{xy} represents shear strain rate associated with lateral deformation.

Principal strain magnitudes and orientations were then calculated to identify regions experiencing maximum compression and extension. The resulting strain rates were converted into nanostrain per year (nstrain/yr) and spatially mapped across the study area to illustrate zones of elevated strain accumulation, particularly along major Himalayan fault systems.

2.8 Slip and Deformation Modeling of Faults

To relate observed surface deformation to subsurface fault behavior, simplified elastic dislocation modeling was applied. Fault geometry parameters, including location, depth, dip, rake, and slip rate, were adopted from published studies of the Main Himalayan Thrust and associated fault systems. Surface displacements generated by fault slip were estimated through forward modeling and compared

with observed GNSS velocity fields to assess fault locking depth, slip deficit, and interseismic coupling.

The modeling approach is based on analytical elastic dislocation solutions in a homogeneous, isotropic half-space following the formulation of Okada (1985). In this framework, surface displacement $u(x,y)$ is expressed as a function of fault geometry and slip parameters:

$$u(x, y) = f(L, W, d, \delta, \lambda, s)$$

where L and W represent fault length and width, d is fault depth, δ is the dip angle, λ is the rake angle, and s denotes the slip rate. The elastic properties of the crust were assumed to be uniform, consistent with common practice in regional-scale crustal deformation studies.

Fault geometry and slip parameters were selected based on established Himalayan megathrust models, particularly those describing the Main Himalayan Thrust in Nepal (Bettinelli et al., 2006; Ader et al., 2012; Stevens & Avouac, 2015). Differences between modeled and observed GNSS velocities were interpreted in terms of interseismic coupling and slip deficit. These deformation patterns were further examined in conjunction with regional seismicity to evaluate the spatial relationship between strain accumulation, fault behavior, and earthquake distribution across the Nepal Himalaya.

2.9 Web-Based Visualization and Integration

All analyses were performed using a web-based GNSS data analysis system developed with standard web technologies. The system was implemented using HTML5 and CSS3 for structure and layout, while JavaScript was used for data loading, computation, and interactive functionality. External libraries for mapping, charting, and scientific visualization were integrated through content delivery networks.

The platform allows users to visualize GNSS velocity fields, station time series, strain rate maps, fault traces, and earthquake distributions directly within a web browser without requiring specialized software. This web-based framework enhances accessibility, supports transparent exploration of results, and promotes reproducibility in crustal deformation analysis.

3. RESULT AND DISCUSSION

3.1 GNSS Velocity Map

GNSS-derived velocity analysis reveals a clear and spatially coherent pattern of crustal motion across Nepal, as shown in Figure 3. All observed GNSS stations exhibit dominant northward and eastward velocity components, reflecting ongoing convergence between the Indian and Eurasian plates. Horizontal velocities across the GNSS network are relatively uniform, ranging from approximately 26.2 to 26.8 mm/yr, which indicates strong interseismic coupling within the Himalayan thrust system. The direction and spatial distribution of these velocity vectors are clearly illustrated in the GNSS velocity field.

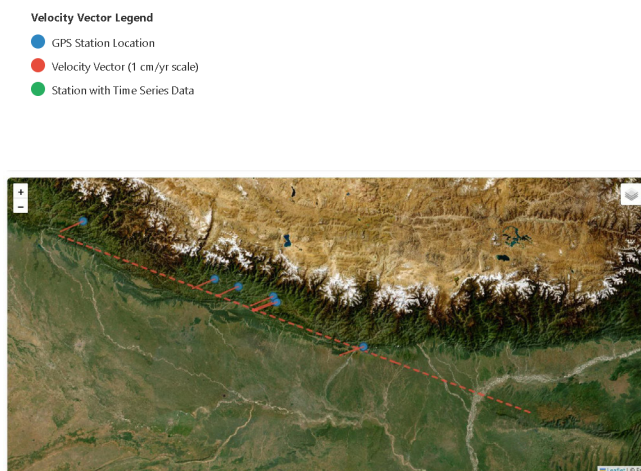


Figure 3:- GNSS-derived horizontal velocity field of the Nepal Himalaya showing present-day crustal motion.

Stations in central Nepal, including CHLM (Kathmandu), KKN4 (Kakani), and NAST (Lalitpur), record eastward velocities of 17.8–18.5 mm/yr and northward velocities of 19.2–20.1 mm/yr, indicating continuous northeastward motion of the Kathmandu Valley region. In contrast, stations located in the northern and western Himalaya, such as SYBC (Syangboche) and JMSM (Jumla), exhibit slightly lower eastward velocity components (15.2–16.8 mm/yr) but higher northward components (20.5–21.3 mm/yr), reflecting spatial variations in strain accommodation along the Himalayan arc.

The eastern Nepal station DMAU (Dharan) shows a pronounced northward velocity of approximately 26.2 mm/yr and an eastward velocity of about 11.6 mm/yr, indicating strong north–northeastward

crustal motion in the eastern Himalayan foreland. This enhanced northward motion suggests significant convergence-related deformation toward the eastern segment of the Himalayan arc, consistent with ongoing interseismic strain accumulation.

Vertical velocity components across the GNSS network are relatively small, ranging from approximately 2.9 to 4.2 mm/yr, indicating slow but persistent crustal uplift. Higher uplift rates are observed at stations in the Higher Himalaya, such as GRHI (Gorkha) and JMSM, consistent with active crustal shortening and ongoing mountain-building processes. The DMAU station exhibits a comparatively lower but continuous vertical uplift rate of approximately 1.9 mm/yr.

3.2 GNSS Time Series Displacement Analysis

Time-series analysis of the CHLM (Kathmandu) GNSS station, presented in Figures 4 and 5, shows a progressive and continuous displacement in the north, east, and vertical components. Figure 5 illustrates the north-component displacement time series of the CHLM station obtained from the Nevada Geodetic Laboratory, which exhibits a strong long-term linear trend superimposed with clear seasonal fluctuations. This pattern indicates steady interseismic crustal deformation associated with ongoing convergence between the Indian and Eurasian plates in the Himalayan region.

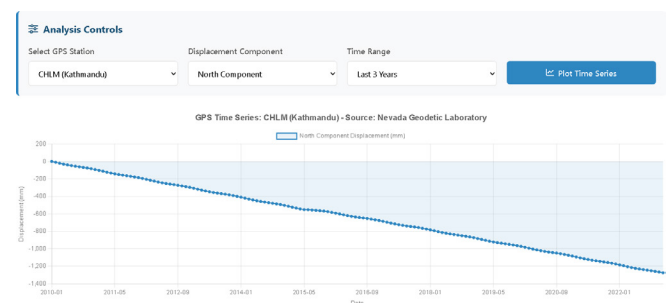


Figure 4:- GNSS displacement time series of the CHLM (Kathmandu) station showing north, east, and vertical components.

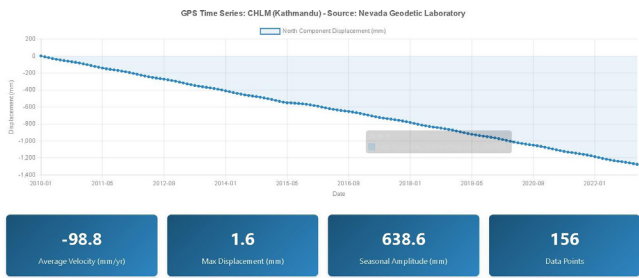


Figure 5:- North-component displacement time series of the CHLM (Kathmandu) GNSS station from the Nevada Geodetic Laboratory, showing a strong long-term linear trend with superimposed seasonal fluctuations indicative of steady interseismic crustal deformation.

The displacement trends are smooth and linear, suggesting that the observed deformation during the study period is predominantly interseismic, with no evidence of significant coseismic offsets. Superimposed seasonal oscillations are evident in several components of the displacement time series and are most likely related to hydrological loading effects associated with the monsoon cycle. At the CHLM station, the maximum seasonal displacement amplitudes are approximately 4.25 mm in the north component, 3.51 mm in the east component, and 1.85 mm in the vertical component.

3.3 Strain Rate Characteristics

Strain-rate analysis indicates that Nepal is dominated by a compressional tectonic regime. GNSS-derived strain rates, including principal strain components, shear strain, and dilatation, are illustrated in Figure 6. The east–west strain component ($\epsilon_{11} = -12.5$ nstrain/yr) and the north–south strain component ($\epsilon_{22} = -15.2$ nstrain/yr) indicate active crustal shortening, with stronger compression in the north–south direction. The shear strain component ($\epsilon_{12} = 8.3$ nstrain/yr) reflects oblique plate convergence accompanied by lateral deformation. In addition, a clockwise rotational rate of approximately 4.2 nrad/yr suggests minor rotational deformation of crustal blocks, particularly in central and eastern Nepal.

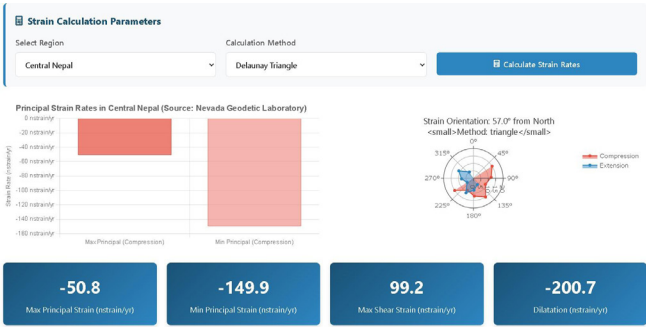


Figure 6:- GNSS-derived strain-rate field of the Nepal Himalaya, showing principal strain, shear strain, and rotational components.

3.4 Seismicity and Deformation Relationship

Seismic activity in Nepal is illustrated in Figure 7, which shows the temporal distribution of earthquakes with magnitudes ≥ 4.0 . Since 2015, a total of 14 earthquakes exceeding this magnitude threshold have been recorded, including a maximum event magnitude of 7.8. The spatial and temporal correspondence between regions of elevated strain accumulation and observed seismicity suggests a strong relationship between crustal deformation patterns and earthquake occurrence in the Himalayan region. Estimated slip deficit rates of approximately 31.5 mm/yr and a fault coupling coefficient of about 0.7 further indicate significant interseismic strain accumulation along major Himalayan thrust faults, highlighting the potential for future large-magnitude seismic events.

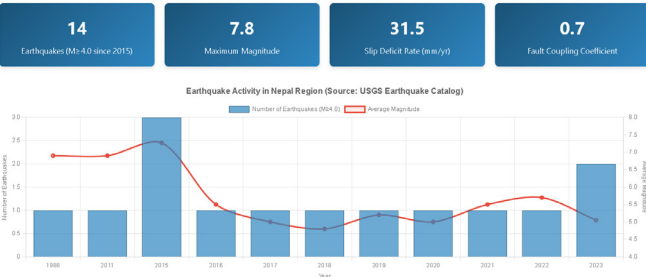


Figure 7:- Temporal distribution of earthquakes ($M \geq 4.0$) in the Nepal region.

4. CONCLUSION

GNSS velocity analysis reveals a consistent and spatially coherent pattern of crustal motion across Nepal, characterized by dominant northward and eastward movements that confirm ongoing con-

vergence between the Indian and Eurasian plates. Relatively uniform horizontal velocity magnitudes ranging from 26.2 to 26.8 mm/yr indicate strong interseismic coupling within the Himalayan thrust system.

Stations in central Nepal, including CHLM (Kathmandu), KKN4 (Kakani), and NAST (Lalitpur), record persistent northeastward motion with eastward velocities of 17.8–18.5 mm/yr and northward velocities of 19.2–20.1 mm/yr. In contrast, northern and western stations such as SYBC (Syangboche) and JMSM (Jumla) exhibit slightly lower eastward velocities (15.2–16.8 mm/yr) but higher northward velocities (20.5–21.3 mm/yr), while the eastern station DMAU (Dharan) shows enhanced northward motion (~26.2 mm/yr), indicating stronger convergence-related deformation in the eastern Himalayan foreland. These spatial variations reflect differences in strain accommodation along the Himalayan arc.

Vertical GNSS velocities are comparatively small (2.9–4.2 mm/yr), indicating slow but continuous crustal uplift. Higher uplift rates in the Higher Himalaya, particularly at GRHI and JMSM, are consistent with active crustal shortening and ongoing mountain-building, whereas DMAU exhibits modest uplift (~1.9 mm/yr). Time-series analysis of the CHLM station further reveals smooth, linear displacement trends in all components, confirming predominantly interseismic deformation with seasonal oscillations likely related to monsoon-driven hydrological loading.

Strain-rate analysis confirms a dominantly compressional tectonic regime across Nepal, with east–west ($\epsilon_{11} = -12.5$ nstrain/yr) and north–south ($\epsilon_{22} = -15.2$ nstrain/yr) shortening, the latter being more pronounced. The presence of shear strain ($\epsilon_{12} = 8.3$ nstrain/yr) and a clockwise rotational rate of approximately 4.2 nrad/yr suggests minor lateral and rotational deformation of crustal blocks.

Seismicity analysis shows a close spatial correspondence between regions of high strain accumulation and earthquake activity. The observed earthquake frequency, maximum magnitude of 7.8, slip deficit rate of approximately 31.5 mm/yr, and fault coupling coefficient of about 0.7 collectively indicate substantial interseismic strain accumulation along the Himalayan thrust system. These findings highlight sustained tectonic loading and emphasize the importance of continuous GNSS monitoring and integrated geodetic–seismic analyses for im-

proving seismic hazard assessment in the Nepal Himalaya.

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